HOSTEL MANAGEMENT SYSTEM

Student Table

```
CREATE TABLE Student (
    student_id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(50),
    usn VARCHAR(20) UNIQUE,
    gender VARCHAR(10),
    department VARCHAR(50),
    year INT,
    phone VARCHAR(15),
    address VARCHAR(100)
);
```

Hostel Table

```
CREATE TABLE Hostel (
    hostel_id INT PRIMARY KEY AUTO_INCREMENT,
    hostel_name VARCHAR(50),
    type VARCHAR(10),   -- Boys/Girls
    total_rooms INT
);
```

Room Table

```
CREATE TABLE Room (
    room_id INT PRIMARY KEY AUTO_INCREMENT,
    hostel_id INT,
    room_number VARCHAR(10),
    capacity INT,
    FOREIGN KEY (hostel_id) REFERENCES Hostel(hostel_id)
```

```
);
```

Allocation Table

```
CREATE TABLE Allocation (
    allocation_id INT PRIMARY KEY AUTO_INCREMENT,
    student_id INT,
    room_id INT,
    join_date DATE,
    FOREIGN KEY (student_id) REFERENCES Student(student_id),
    FOREIGN KEY (room_id) REFERENCES Room(room_id)
);
```

Fee Table

```
CREATE TABLE Fee (
    fee_id INT PRIMARY KEY AUTO_INCREMENT,
    student_id INT,
    amount DECIMAL(10,2),
    status VARCHAR(20),  -- Paid / Pending
    payment_date DATE,
    FOREIGN KEY (student_id) REFERENCES Student(student_id)
);
```

Warden Table

```
CREATE TABLE Warden (
    warden_id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(50),
    phone VARCHAR(15),
    hostel_id INT,
    FOREIGN KEY (hostel_id) REFERENCES Hostel(hostel_id)
);
```

INSERT QUERIES

Insert Student

INSERT INTO Student (name, usn, gender, department, year, phone, address)

VALUES ('Ananya', '1VT21EC001', 'Female', 'ECE', 3, '9876543210', 'Bangalore');

Insert Hostel

INSERT INTO Hostel (hostel_name, type, total_rooms)

VALUES ('Ganga Hostel', 'Girls', 100);

Insert Room

INSERT INTO Room (hostel_id, room_number, capacity)

VALUES (1, 'G101', 3);

Allocate Room

INSERT INTO Allocation (student_id, room_id, join_date)

VALUES (1, 1, '2025-01-01');

Fee Payment

INSERT INTO Fee (student_id, amount, status, payment_date)

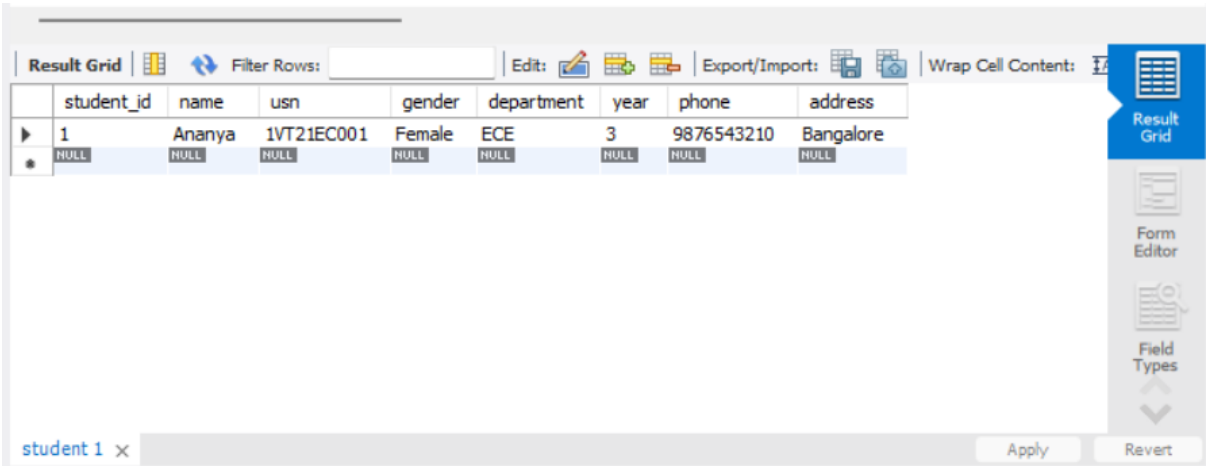
VALUES (1, 45000, 'Paid', '2025-01-10');

---

4. SELECT QUERIES

View All Students

SELECT * FROM Student;



Students in a Particular Department

```
SELECT name, usn

FROM Student

WHERE department = 'ECE';
```

Students Staying in Hostel

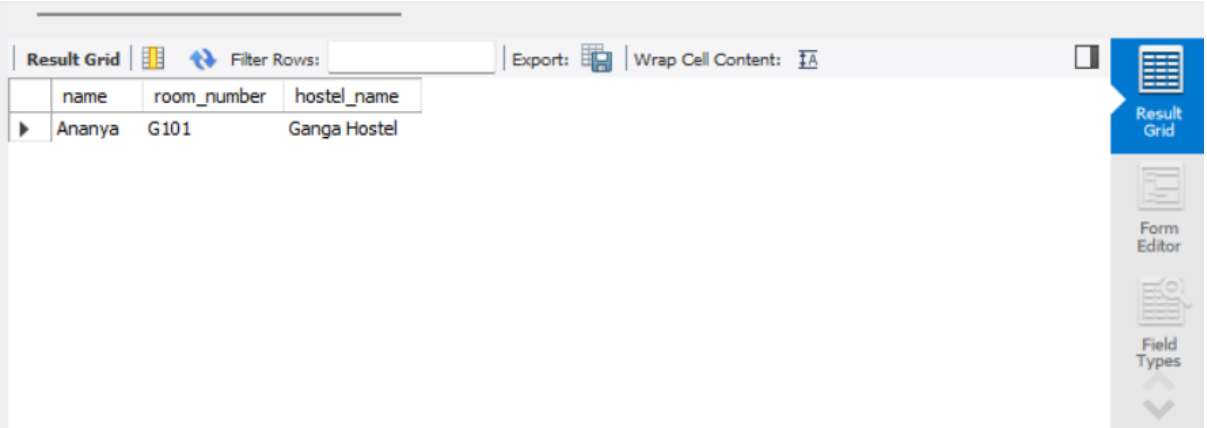
```
SELECT s.name, r.room_number, h.hostel_name

FROM Student s

JOIN Allocation a ON s.student_id = a.student_id

JOIN Room r ON a.room_id = r.room_id

JOIN Hostel h ON r.hostel_id = h.hostel_id;
```



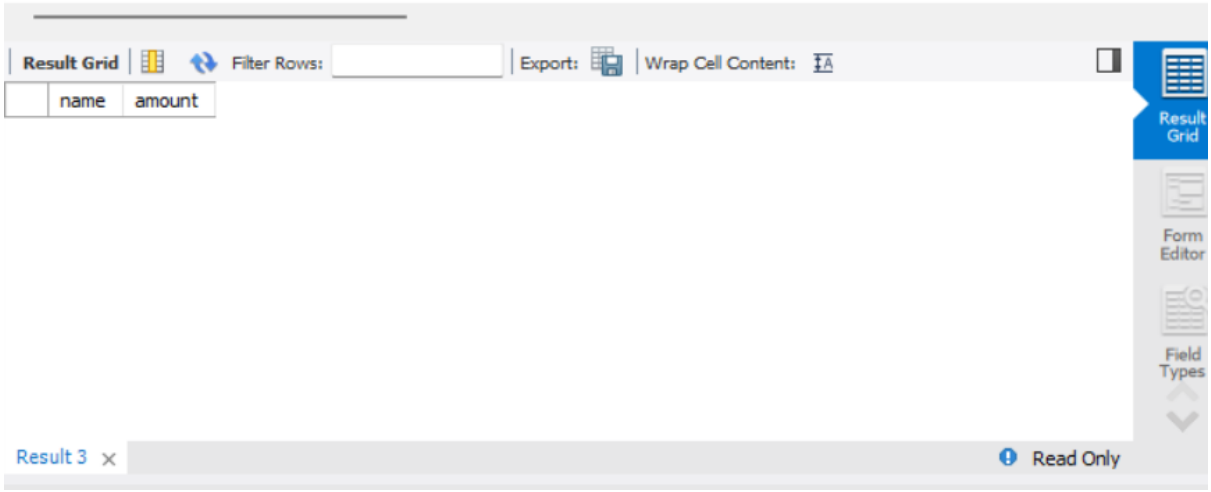
Students with Pending Fees

```
SELECT s.name, f.amount

FROM Student s

JOIN Fee f ON s.student_id = f.student_id

WHERE f.status = 'Pending';
```



Total Students in Each Hostel

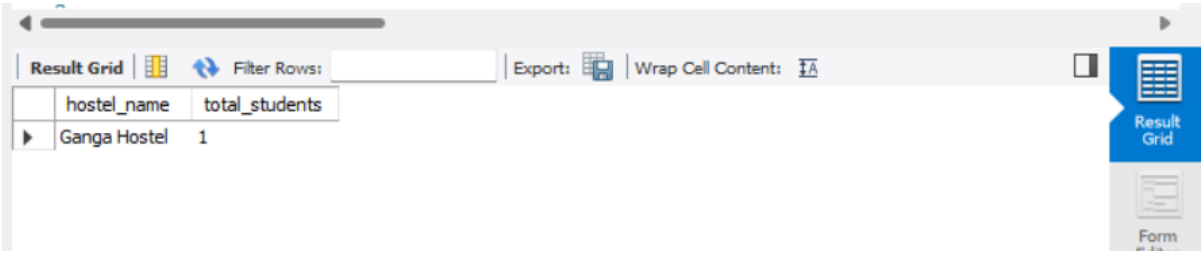
SELECT h.hostel_name, COUNT(a.student_id) AS total_students

FROM Hostel h

JOIN Room r ON h.hostel_id = r.hostel_id

JOIN Allocation a ON r.room_id = a.room_id

GROUP BY h.hostel_name;



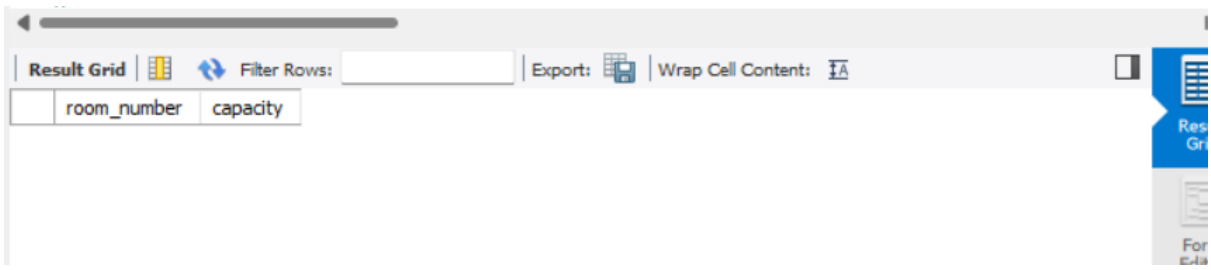
Available Rooms

SELECT r.room_number, r.capacity

FROM Room r

LEFT JOIN Allocation a ON r.room_id = a.room_id

WHERE a.room_id IS NULL;



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5. UPDATE QUERIES

Update Student Phone Number

```
UPDATE Student
SET phone = '9999999999'
WHERE usn = '1VT21EC001';
```

Update Fee Status

```
UPDATE Fee
SET status = 'Paid', payment_date = CURDATE()
WHERE student_id = 1;
```

---

6. DELETE QUERIES

Remove a Student

```
DELETE FROM Student
WHERE student_id = 1;
```

Remove Room Allocation

```
DELETE FROM Allocation
WHERE student_id = 1;
```

---

7. AGGREGATE & FUNCTIONS

Total Fee Collected

```
SELECT SUM(amount) AS total_fee
FROM Fee
WHERE status = 'Paid';
```

Number of Rooms in Each Hostel

```
SELECT hostel_id, COUNT(room_id) AS total_rooms
FROM Room
GROUP BY hostel_id;
```

---

8. SUBQUERIES

Students Who Have Not Paid Fees

```
SELECT name
FROM Student
WHERE student_id IN (
   SELECT student_id
   FROM Fee
   WHERE status = 'Pending'
);
```

---

9. VIEWS

```
CREATE VIEW HostelStudents AS
SELECT s.name, h.hostel_name, r.room_number
FROM Student s
JOIN Allocation a ON s.student_id = a.student_id
JOIN Room r ON a.room_id = r.room_id
JOIN Hostel h ON r.hostel_id = h.hostel_id;
```